

STATISTICAL ANALYSIS SUMMARY

1. Introduction

Statistical analysis was conducted during week 3 to investigate whether the patterns observed during exploratory data analysis were supported by statistical evidence. The analysis focused on relationships between applicant characteristics and the loan approval outcome, Loan_Status. A significance level of 5% ($\alpha = 0.05$) was used to interpret the hypothesis tests. A p-value below 0.05 was considered evidence of a statistically significant relationship or difference. The statistical analysis included correlation analysis, chi-square tests for categorical variables and the Mann-Whitney U test for group comparisons.

2. Correlation analysis

Correlation analysis was performed to examine the strength and direction of relationships between numerical variables. The correlation results showed several notable relationships.

Variable Pair	Correlation
ApplicantIncome – TotalIncome	0.8930
LoanAmount – TotalIncome	0.6246
ApplicantIncome – LoanAmount	0.5709
TotalIncome – DebtToIncomeRatio	-0.3882

Interpretation

The strongest relationship was observed between ApplicantIncome and TotalIncome ($r = 0.8930$). This strong positive relationship is expected because TotalIncome incorporates applicant and coapplicant income. A moderate positive relationship was observed between LoanAmount and TotalIncome ($r = 0.6246$), indicating that higher total income was generally associated with larger requested loan amounts. ApplicantIncome and LoanAmount also showed a moderate positive relationship ($r = 0.5709$). The relationship between TotalIncome and DebtToIncomeRatio was negative ($r = -0.3882$), indicating that higher income was generally associated with lower values of the debt-to-income measure.

3. Chi-Square analysis

Chi-square tests of independence were used to examine whether categorical variables were statistically associated with Loan_Status. The results identified statistically significant relationships for several variables.

Variable	p-value	Result
----------	---------	--------

Married	0.034394	Significant
Education	0.043100	Significant
Property_Area	0.002136	Significant

3.1 Married and loan status

The chi-square test for Married produced a p-value of 0.034394. Since the p-value is below 0.05, there is evidence of a statistically significant association between marital status and loan approval status in this dataset. This suggests that loan approval outcomes were different across the observed marital status groups.

3.2 Education and loan status

The chi-square test for Education produced a p-value of 0.043100. Because the p-value is below 0.05, the relationship between education status and Loan_Status was statistically significant at the 5% level. This indicates that the distribution of loan approval outcomes differed between the education groups represented in the dataset.

3.3 Property area and loan status

The chi-square test for Property_Area produced a p-value of 0.002136. This is below the 0.05 significance threshold, providing evidence of a statistically significant association between property area and loan approval status. Among the categorical variables tested, Property_Area therefore provided particularly strong statistical evidence of an association with Loan_Status.

4. Other categorical variables

Not all evaluated categorical variables demonstrated statistically significant relationships with Loan_Status.

The following variables had p-values above 0.05:

Variable	p-value	Interpretation
Gender	0.708653	Not statistically significant
Dependents	0.367851	Not statistically significant
Self_Employed	1.000000	Not statistically significant
IncomeBand	0.889746	Not statistically significant
FamilySizeGroup	0.784945	Not statistically significant

These results indicate that there was insufficient statistical evidence to conclude that these variables were associated with Loan_Status at the 5% significance level. However, lack of statistical significance does not

automatically mean that a variable has no predictive value. These variables were therefore considered alongside exploratory findings, feature relationships and modelling requirements rather than being removed solely on the basis of their individual p-values.

5. CreditRiskCategory analysis

CreditRiskCategory demonstrated a strong statistical association with Loan_Status.

The chi-square test produced:

- $\chi^2 = 180.0665$
- $p < 0.001$

The result provides strong evidence of an association between CreditRiskCategory and loan approval status. However, CreditRiskCategory was not retained in the final modelling dataset. The feature was derived from Credit_History, which was already available as an original predictor. Including both variables would therefore introduce redundant information. For this reason, CreditRiskCategory was removed while Credit_History was retained.

6. Mann-Whitney U test

A Mann-Whitney U test was conducted to compare the distributions of the two groups being examined.

The test produced:

- U statistic = 38,267.0
- p-value = 0.576442

Since the p-value is greater than 0.05, the result was not statistically significant. Therefore, there was insufficient evidence to conclude that the two groups differed significantly for the variable tested. This finding demonstrates the importance of using statistical evidence when interpreting apparent differences observed during exploratory analysis.

7. Overall statistical findings

The statistical analysis identified several important relationships within the loan prediction dataset.

The main findings were:

1. Married showed a statistically significant association with Loan_Status,
2. Education showed a statistically significant association with Loan_Status,
3. Property_Area showed a statistically significant association with Loan_Status,
4. CreditRiskCategory showed a very strong statistical association with Loan_Status but was removed because it duplicated Credit_History,

5. ApplicantIncome and TotalIncome showed a very strong positive correlation,
6. LoanAmount showed positive relationships with both ApplicantIncome and TotalIncome,
7. The Mann-Whitney U test did not provide evidence of a statistically significant difference between the groups tested.

8. Implications for feature selection

Statistical findings were used as supporting evidence during the feature evaluation and dataset refinement process. Variables with statistically significant relationships were retained as potentially informative predictors. However, statistical significance was not treated as the only criterion for feature selection.

Feature selection also considered:

- Business relevance
- Redundancy
- Feature construction
- Correlation between predictors
- Interpretability
- Potential usefulness for machine-learning models

This approach helped ensure that the final modelling dataset was not created solely from statistical significance tests but instead reflected the broader analytical findings from week 3.

9. Conclusion

The statistical analysis provided additional evidence for several patterns identified during exploratory data analysis. Significant associations were observed between Loan_Status and Married, Education and Property_Area, while strong correlation was observed among several financial variables. The analysis also identified redundant information through the evaluation of CreditRiskCategory. Removing this feature while retaining Credit_History helped reduce unnecessary duplication in the final modelling dataset. Overall, the statistical analysis strengthened the evidence base used to refine the dataset and prepare it for predictive modelling in week 4.